

Part 1 of 4

Precision-Guided Munitions: Guidance Techniques



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Precision has always been one of the five most important attributes of a weapon, with operational range, striking power, volume of fire and portability being the other four. The state-of-the-art precision-guided munitions (PGMs) combine all these attributes to make them a potent force multiplier in the contemporary battlefield. Emergence and subsequent maturity of PGMs, made possible mainly due to advances in electronics, optoelectronics and optics, is one of the most significant developments of modern warfare. Let us discuss the various guidance techniques employed in this class of weapons.

PGMs, also called smart munitions, belong to the group of advanced fire-power munitions which mainly include projectiles fired from land or ship based military platforms, surface-to-air and air-to-air missiles and aerially-delivered bombs. These weapons employ one or more guidance techniques to hit the target more precisely with minimised collateral damage than would be possible with conventional unguided weapons. Launched from a variety of military platforms, including land vehicles, aircraft, ships and submarines, these weapons exemplify the concept of a low-cost threat, forcing a high-cost and complicated defence mechanism.

Though the concept of PGMs was first envisaged during World War I, the scientific and technological capability of that time did not allow it to become a practical reality. This weapon arrived on the battlefield in a rudimentary yet significant form during World War II. It was the success and experience gained during the Korean and Vietnam conflicts, and then the Gulf war, that established the efficacy of precision-strike weapons.



Fig. 1: AGM-88 HARM ARM



Fig. 2: MM40 EXOCET ARM

This class of weapons has come a long way from the era of World War II, when the guided-weapon was characterised by a circular error probable (CEP) of about 1000 metres to the state-of-the-art precision-strike weapon with CEP of less than a metre.

New guidance mechanisms are being developed to further enhance the precision of these weapons. More than one guidance technologies are being employed to make them all-weather weapons and eliminate the possibility of a mis-hit. Use of GPS/INS in conjunction with other guidance technologies, such as radar or electro-optic guidance, is an example.

Joint direct attack munitions (JDAM) is an important manifestation of the use of multiple-guidance technologies. The capability of a modern precision weapon has progressed from targeting a specific build-